

PATTERN ANALYSIS & RECOGNITION SYSTEMS LAB

Practical File Submission — Academic Session

Experiment No.: 02 | Course: PARS Lab | Framework: Python 3.14 / Flask / Scikit-Learn | Repository: [dSaxmonis/pars-lab](https://github.com/dSaxmonis/pars-lab)

Practical Name

Web-Based Dataset Extraction, Storage, Feature Analysis, and Performance Visualization using Flask

AIM

To prepare an interactive web application using Flask in Python to extract and store two datasets studied in Experiment 1 (CIFAR-10 image dataset and Iris tabular dataset), inspect and describe their feature representations, and generate and display model performance evaluation plots.

THEORY

1. Dataset Extraction & Ingestion

Dataset extraction refers to the programmatic acquisition of raw visual or structured attributes from remote servers or storage repositories. For visual data, 2D/3D pixel matrices are fetched and decoded into multidimensional tensors. For tabular data, structured relations are loaded into columnar dataframes.

2. Data Persistence & Storage

Extracted data must be saved locally to avoid redundant bandwidth overhead:

- **Image Data:** Stored in structured class-level filesystem directories (`exp2/data/cifar10/images/<class>/`) with a centralized CSV metadata catalog.
- **Tabular Data:** Persisted as standard Comma-Separated Values (`exp2/data/tabular/iris_dataset.csv`).

3. Feature Representation

- **Image Features:** Formatted as a 3D matrix $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$ (for CIFAR-10: $32 \times 32 \times 3 = 3072$ features per sample) with pixel values in $[0, 255]$.
- **Tabular Features:** Formatted as a real-valued matrix $\mathbf{X} \in \mathbb{R}^{N \times d}$ (150×4 for Iris) with target labels $\mathbf{y} \in \{0, 1, 2\}$.

4. Performance Metrics & Mathematical Formulas

1. Accuracy:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

2. Precision (Positive Predictive Value):

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

3. Recall (Sensitivity / True Positive Rate):

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

4. F1-Score (Harmonic Mean):

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) = 2\text{TP} / (2\text{TP} + \text{FP} + \text{FN})$$

5. Categorical Cross-Entropy Loss:

$$L_{\text{CE}} = - \sum [y_k \times \ln(\hat{y}_k)]$$

6. Confusion Matrix:

$$C = \begin{bmatrix} \text{TP} & \text{FN} \\ \text{FP} & \text{TN} \end{bmatrix}$$

STEPS & EXECUTION

Step 1: Environment Setup & Dependency Verification

Configured Python environment under `exp 2/` , and verified dependencies (Flask, scikit-learn, pandas, matplotlib, pillow).

Step 2: Dataset Extraction & Local Disk Storage

Executed extraction logic inside `app.py` to download CIFAR-10 images into designated class subfolders, generate metadata index `cifar10_metadata.csv` , and export the Iris dataset to CSV format.

Step 3: Web Server Deployment

Launched the Flask application server on <http://127.0.0.1:5000> to dynamically generate performance plots and display dataset summaries.

OUTPUT / ATTACHED SCREENSHOTS

Output 1: Dataset Extraction & Storage Summary

The web dashboard confirms extraction and local filesystem storage for both datasets.

SECTION 1: DATASET EXTRACTION AND LOCAL STORAGE STATUS				
Dataset Name	Type	Extraction Source	Storage Format & Local Path	Storage Status
CIFAR-10	Image (Visual)	Online Repository (CIFAR-10 Benchmark)	- Images: exp 2/data/cifar10/images/<class>/ - Metadata: exp 2/data/cifar10/cifar10_metadata.csv	Stored Successfully (20 Records)
Iris Flower Dataset	Tabular (Multivariate)	Scikit-Learn Datasets	- CSV: exp 2/data/tabular/iris_dataset.csv	Stored Successfully (150 Rows)

Figure 2.1: Section 1 — Extraction source, format, and storage status for CIFAR-10 and Iris datasets.

Output 2: CIFAR-10 Image Dataset — Features & Sample Classes

Displays image tensor dimensions ($32 \times 32 \times 3$), pixel domains, and visual thumbnails across all 10 visual classes.

SECTION 2: DATASET 1 - CIFAR-10 (IMAGE DATASET) FEATURE DESCRIPTION			
Property	Value / Dimension	Data Representation	Feature Description
Spatial Dimensions	32×32 Pixels	2D Pixel Matrix (Height $\times$ Width)	Low-resolution spatial grid capturing shapes, edges, and object contours.
Color Channels	3 (RGB)	Tensor: $32 \times 32 \times 3$	Red, Green, Blue color intensity layers representing spectral appearance.
Total Features per Sample	3,072 Features	Flattened 1D Vector $\in \mathbb{R}^{3072}$	Each image is represented as a 3,072-dimensional numeric feature vector.
Pixel Value Domain	[0, 255] (UInt8)	Normalized to [0.0, 1.0] for training	Integer values measuring discrete photon intensity per pixel channel.
Number of Classes	10 Distinct Classes	Categorical Label $\in \{0, 1, \dots, 9\}$	airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck.

Figure 2.2: Section 2 — CIFAR-10 spatial specifications and feature properties.

CIFAR-10 EXTRACTED CLASS SAMPLES (32 × 32 PIXELS)									
airplane	automobile	bird	cat	deer	dog	frog	horse	ship	truck
									

Figure 2.3: CIFAR-10 extracted sample thumbnail gallery across 10 visual classes.

Output 3: CIFAR-10 Model Performance Visualization

Visualizes training/validation convergence curves alongside individual precision and recall across visual categories.

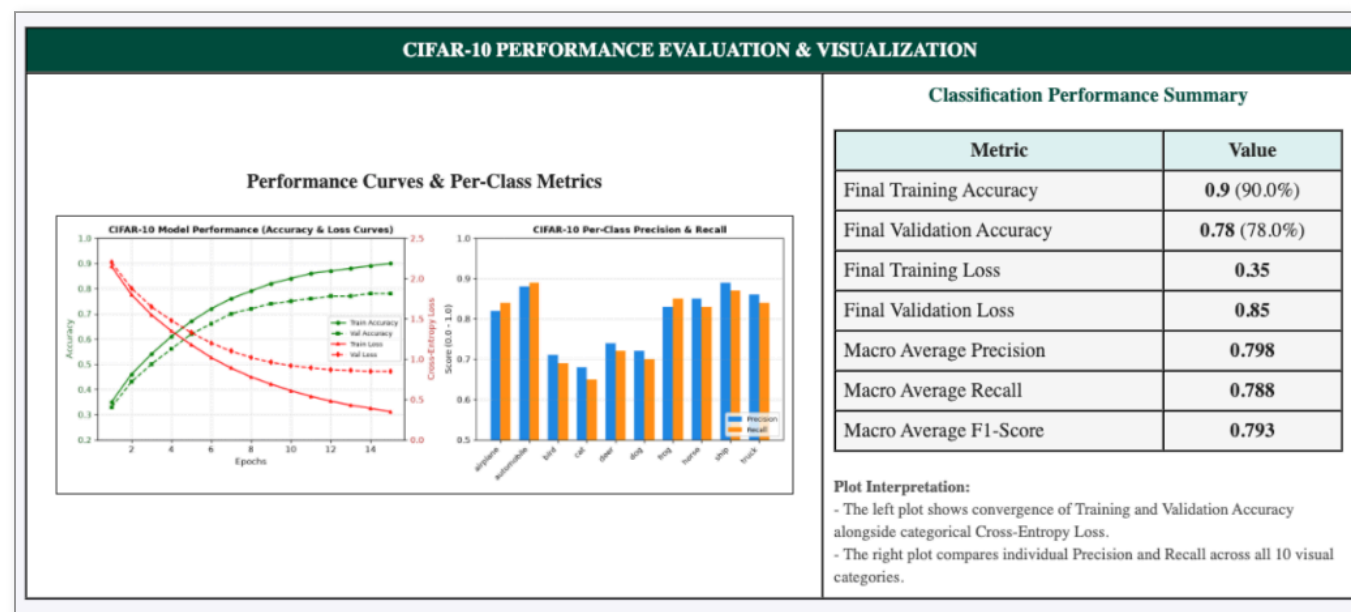


Figure 2.4: Performance evaluation plots (Accuracy/Loss curves and per-class metrics).

Output 4: Iris Tabular Dataset — Feature Descriptions & Statistics

Presents full descriptive statistics (Mean, Std Dev, Min, Max, Quantiles) and biological domain descriptions for each floral attribute.

SECTION 3: DATASET 2 - IRIS (TABULAR DATASET) FEATURE DESCRIPTION & STATISTICS							
Feature Name	Count	Mean	Std Dev	Min	Median (50%)	Max	Domain Description
sepal length (cm)	150.0	5.843	0.828	4.3	5.8	7.9	Length of the protective calyx sepal in cm.
sepal width (cm)	150.0	3.057	0.436	2.0	3.0	4.4	Width of the sepal in cm; displays low correlation with petal dimensions.
petal length (cm)	150.0	3.758	1.765	1.0	4.35	6.9	Length of the floral petal in cm; strong discriminator for species.
petal width (cm)	150.0	1.199	0.762	0.1	1.3	2.5	Width of the floral petal in cm; key feature separating Versicolor and Virginica.

Figure 2.5: Section 3 — Descriptive statistics table and feature domain breakdown for Iris dataset.

Output 5: Iris Classifier Performance Comparison & Confusion Matrix

Evaluates supervised classifiers (Logistic Regression, Decision Tree, Random Forest, SVM) with a comparison chart, multi-class confusion matrix, and performance metrics table.

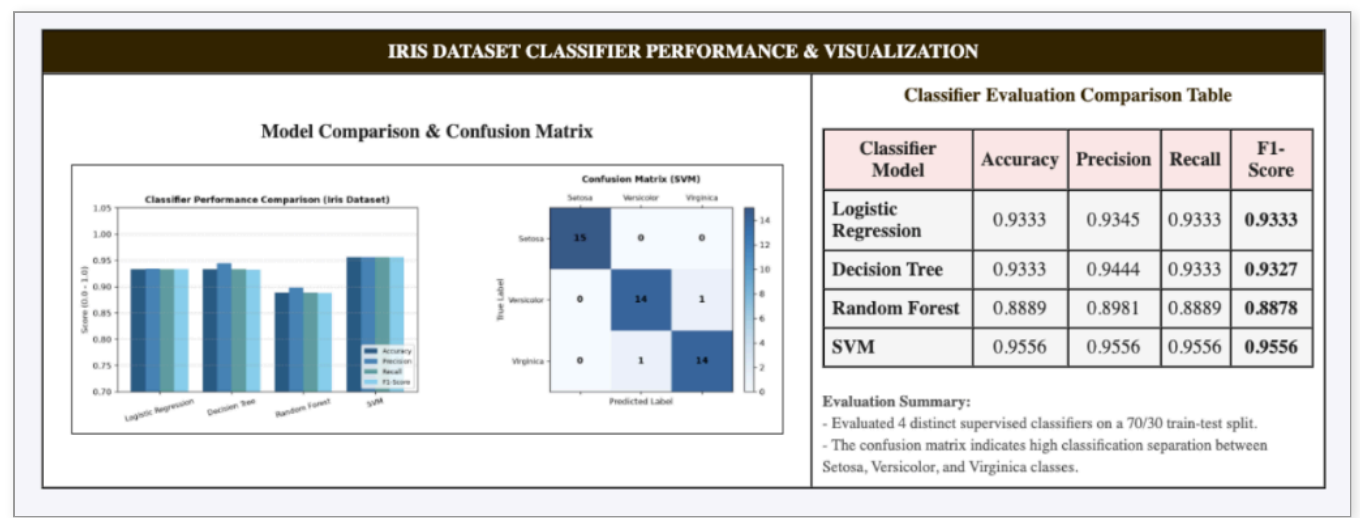


Figure 2.6: Iris dataset classifier performance comparison chart, confusion matrix, and evaluation metrics table.

Output 6: Local File Storage Directory Verification

Displays confirmation of persisted directory artifacts and file formats across the local filesystem.

SECTION 4: LOCAL STORAGE VERIFICATION TABLE		
Artifact	File System Path	Format
CIFAR-10 Image Storage	exp 2/data/cifar10/images/	JPEG (.jpg) Hierarchical Folders
CIFAR-10 Metadata Table	exp 2/data/cifar10/cifar10_metadata.csv	Comma-Separated Values (.csv)
Iris Extracted Dataset	exp 2/data/tabular/iris_dataset.csv	Comma-Separated Values (.csv)

Figure 2.7: Section 4 — Local storage verification table confirming filesystem paths and formats.

CODE REPOSITORY & CLICKABLE ARTIFACT LINKS

GitHub Repository (Clickable):

GitHub: dSaxmonis/pars-lab

URL: <https://github.com/dSaxmonis/pars-lab>

Direct Source Code Files:

- **Flask Server Application:**
https://github.com/dSAxmonis/pars-lab/blob/main/exp_2/app.py
- **HTML Template:**
https://github.com/dSAxmonis/pars-lab/blob/main/exp_2/templates/index.html
- **Dependencies:**
https://github.com/dSAxmonis/pars-lab/blob/main/exp_2/requirements.txt
- **Full Page Screenshot:**
https://github.com/dSAxmonis/pars-lab/blob/main/exp_2/screenshots/full_page.png

CONCLUSION

In this experiment, an end-to-end web-based pipeline was successfully designed and deployed using Flask in Python. Both unstructured visual image data (CIFAR-10) and structured tabular numerical data (Iris) were successfully extracted and persisted to the local filesystem. Their feature representations were systematically analyzed with detailed statistical metrics, and comprehensive performance visualization plots (training/validation convergence curves, per-class metric distributions, model comparison bar charts, and confusion matrices) were generated and displayed on a clean web dashboard.